Relational Retrieval: Leveraging Known-Novel Interactions for Generalized Category Discovery

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Abstract

In this study, we tackle Generalized Category Discovery (GCD) via a Relational Retrieval perspective, explicitly coupling labeled and unlabeled data through bidirectional knowledge transfer. While existing methods treat these sources separately, missing valuable interaction opportunities, we propose Relational Pattern Consistency (RPC) that enables mutual enhancement. RPC employs One-vs-All classifiers for soft ID/OOD decomposition, then introduces two mechanisms: (i) for known-class preservation, we transfer semantic behavioral alignment; (ii) for category discovery, we leverage the insight that samples from the same category maintain invariant relationships with known-class prototypes, transforming unreliable pseudo-labeling into well-defined relational pattern matching. This bidirectional design allows labeled data to guide unlabeled learning while discovering novel categories through their collective relational signatures. Extensive experiments demonstrate RPC achieves state-of-the-art performance on both generic and fine-grained benchmarks. Code will be released upon acceptance.

CCS Concepts

• **Computing methodologies → Visual content-based indexing and retrieval**.

Keywords

Generalized category discovery, semi-supervised learning, contrastive learning.

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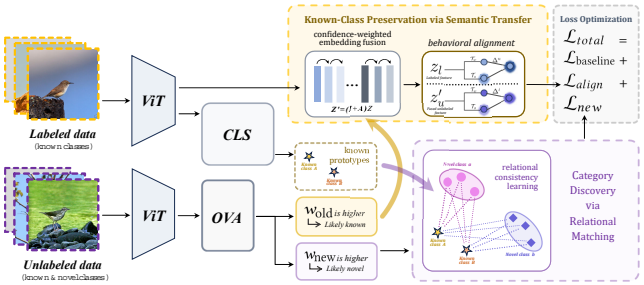

Figure 1: Overview of RPC. Soft ID/OOD decomposition via OVA classifiers enables bidirectional knowledge transfer: embedding fusion for known-class preservation and relational matching for category discovery.

1 Introduction

Generalized Category Discovery (GCD) [? ?] extends traditional classification to open-world scenarios by jointly recognizing known classes and discovering novel classes in unlabeled data. This problem has been explored across various tasks, including natural images [? ? ? ? ? ? ?] and specialized applications such as medical imaging and healthcare areas [? ? ?], industrial scenarios [?], face recognition [? ?], information retrieval [?], etc.

Recent advances in GCD [? ? ? ? ? ? ? ? ?] have made significant progress through various strategies, including parametric classifier learning [?], spatial prompt tuning [?], textual information integration [? ?] and soft ID/OOD decomposition [?], among others. However, these approaches fundamentally treat labeled and unlabeled data as separate learning streams—labeled data undergoes supervised learning while unlabeled data relies on self-supervised signals. This separation overlooks a critical insight: labeled and unlabeled data can mutually enhance each other through structured interactions, where labeled data provides semantic guidance while unlabeled data offers distributional knowledge. We identify two key limitations of this separation. First, unlabeled known-class samples rely solely on self-supervision for representation [?], despite labeled samples of the same classes having already captured highly discriminative features—wasting valuable semantic knowledge. Second, category discovery proceeds without considering relational patterns—samples from the same novel category, while semantically distinct from known classes, maintain consistent relationships with known-class prototypes that serve as powerful grouping signals.

In this study, we propose Relational Pattern Consistency (RPC), a novel framework that explicitly couples labeled and unlabeled data through bidirectional knowledge transfer as shown in Fig. 1. RPC leverages soft ID/OOD decomposition from One-vs-All classifiers to enable targeted interactions: (i) for known-class preservation, we transfer semantic patterns from labeled to unlabeled data through confidence-weighted embedding fusion and enforce behavioral alignment under augmentation [?], ensuring unlabeled known-class samples inherit the robust representations from labeled data; (ii) for category discovery, we introduce relational consistency—the principle that samples from the same category maintain invariant relationships with known-class prototype. This transforms discovery from ill-posed clustering into well-defined pattern matching within the known-class reference frame. We view this process as *relational retrieval*: category identity is inferred by matching a sample's relational signature against the known-class reference frame rather than by direct feature comparison.

In summary, our contributions are three-fold: (i) We propose RPC, the first GCD framework that couples labeled and unlabeled data through bidirectional knowledge transfer. Unlike existing methods that maintain separate learning streams, RPC employs soft ID/OOD decomposition to enable targeted interactions. (ii) We introduce confidence-weighted embedding fusion with behavioral alignment for known-class preservation, and relational consistency learning that discovers novel categories through their invariant relationships with known-class prototypes, going beyond unreliable pseudo-labels. (iii) Extensive experiments demonstrate that RPC achieves state-of-the-art performance on both generic and fine-grained benchmarks, with consistent improvements in known and novel class accuracy validating our bidirectional transfer approach.

2 Problem Setting

Let $\mathcal{D}$ denote the dataset, split into a labeled part $\mathcal{D}_L = \{(\mathbf{x}_i^l, y_i^l)\} \subset \mathcal{X} \times \mathcal{Y}_l$ and an unlabeled part $\mathcal{D}_U = \{\mathbf{x}_i^u\} \subset \mathcal{X}$, where each $\mathbf{x}_i^u$ is associated with a hidden ground-truth label $\hat{y}_i \in \mathcal{Y}_u$ used only for evaluation. The unlabeled pool may contain categories absent from the labeled subset, i.e., $\mathcal{Y}_l \subset \mathcal{Y}_u$. The learning objective is to assign every instance in $\mathcal{D}_U$ to a semantic category. Following [?], we assume the total number of categories in the unlabeled set is known a priori, and write $K = |\mathcal{Y}_u|$.

3 Method

3.1 Baseline for GCD

In this study, our model builds on SimGCD [?] for representation learning and parametric classification, while incorporating DebGCD's [?] One-vs-All (OVA) decomposition to identify whether unlabeled samples belong to known or novel classes. This combination enables effective feature learning through SimGCD while OVA decomposition provides reliable soft weights for our bidirectional knowledge transfer.

3.1.1 Representation Learning. The objective of representation learning is to obtain discriminative features, which allows the classifier effectively classify all the categories. To do so, a pre-trained feature extractor $f(\cdot)$ is fine-tuned with two contrastive losses. Given two random augmentations $\hat{\mathbf{x}}_i$ and $\tilde{\mathbf{x}}_i$ of an image $\mathbf{x}_i$ in a training mini-batch $\mathcal{B}$ that contains both labeled and unlabeled data, the self-supervised contrastive loss is formulated as:

$$\mathcal{L}_{\text{rep}}^u = \frac{1}{|\mathcal{B}|} \sum_{\mathbf{x}_i \in \mathcal{B}} -\log \frac{\exp(\hat{\mathbf{z}}_i^\top \tilde{\mathbf{z}}_i / \tau_u)}{\sum_{\mathbf{x}_j \in \mathcal{B}} \exp(\hat{\mathbf{z}}_i^\top \tilde{\mathbf{z}}_j / \tau_u)}, \tag{1}$$

where $\mathbf{z}_i = g(f(\mathbf{x}_i))$ is the projected feature for contrastive learning, $g(\cdot)$ is the projection head, and τ_u is a temperature value. A supervised contrastive loss is similarly defined by:

$$\mathcal{L}_{\text{rep}}^s = \frac{1}{|\mathcal{B}_l|} \sum_{\mathbf{x}_i \in \mathcal{B}_l} \frac{1}{|\mathcal{N}_i|} \sum_{p \in \mathcal{N}_i} -\log \frac{\exp(\hat{\mathbf{z}}_i^\top \tilde{\mathbf{z}}_p / \tau_s)}{\sum_{n \neq i} \exp(\hat{\mathbf{z}}_i^\top \tilde{\mathbf{z}}_n / \tau_s)}, \tag{2}$$

where $\mathcal{N}_i$ is the indexes of images in the mini-batch that have the same label to $\mathbf{x}_i$, $\mathcal{B}_l$ is the mini-batch of labeled training data and τ_s is a temperature value. The two losses are combined to learn the representation: $\mathcal{L}_{\text{rep}} = (1-\lambda)\mathcal{L}_{\text{rep}}^u + \lambda \mathcal{L}_{\text{rep}}^s$, where λ is a hyperparameter.

3.1.2 Classifier Learning. Following SimGCD [?], we employ a parametric classifier with learnable prototypes $\mathcal{T} = \{\mathbf{t}_1, \mathbf{t}_2, \ldots, \mathbf{t}_K\}$, where $K = |\mathcal{Y}_u|$ is assumed known a priori. During training, the soft label $\hat{p}_i^k$ for each augmented view $\hat{\mathbf{x}}_i$ is computed via softmax over cosine similarities between features and prototypes:

$$\hat{p}_i^k = \frac{\exp\left((\hat{\mathbf{h}}_i / \|\hat{\mathbf{h}}_i\|_2)^\top (\mathbf{t}_k / \|\mathbf{t}_k\|_2) / \tau_s\right)}{\sum_j \exp\left((\hat{\mathbf{h}}_i / \|\hat{\mathbf{h}}_i\|_2)^\top (\mathbf{t}_j / \|\mathbf{t}_j\|_2) / \tau_s\right)}, \tag{3}$$

where $\hat{\mathbf{h}}_i = f(\hat{\mathbf{x}}_i)$. Similarly computing $\tilde{\mathbf{p}}_i$ for $\tilde{\mathbf{x}}_i$, the classifier losses become:

$$\mathcal{L}_{\text{cls}}^s = \frac{1}{|\mathcal{B}_l|} \sum_{\mathbf{x}_i \in \mathcal{B}_l} \ell_{CE}(\mathbf{y}(\mathbf{x}_i), \hat{\mathbf{p}}_i), \quad \mathcal{L}_{\text{cls}}^u = \frac{1}{|\mathcal{B}|} \sum_{\mathbf{x}_i \in \mathcal{B}} \ell_{CE}(\tilde{\mathbf{p}}_i, \hat{\mathbf{p}}_i), \tag{4}$$

where $\mathbf{y}_i$ is the ground truth label for the labeled data point $\mathbf{x}_i$, ℓ_{CE} is the cross-entropy loss, and $H(\mathbf{p}) = -\sum \mathbf{p} \log \mathbf{p}$ regularizes the mean prediction $\mathbf{p} = \frac{1}{2|\mathcal{B}|} \sum_{\mathbf{x}_i \in \mathcal{B}} (\hat{\mathbf{p}}_i + \tilde{\mathbf{p}}_i)$ in a mini-batch. The final classifier loss is given by $\mathcal{L}_{\text{cls}} = (1-\lambda)\mathcal{L}_{\text{cls}}^u + \lambda \mathcal{L}_{\text{cls}}^s - \epsilon H(\mathbf{p})$. Combining the losses of the representation learning and the classifier learning, the overall loss of the baseline SimGCD is formulated by: $\mathcal{L}_{\text{baseline}} = \mathcal{L}_{\text{rep}} + \mathcal{L}_{\text{cls}}$.

3.1.3 One-vs-All Decomposition. Following DebGCD [?], we employ One-vs-All (OVA) classifiers for soft ID/OOD decomposition. For each known class $c \in \mathcal{Y}_l$, we pre-train a binary classifier $h_c : \mathbb{R}^d \to [0, 1]$ to distinguish class c from all others. For each unlabeled sample, we aggregate the per-class confidences into a single ID score $s_{ID}(\mathbf{x}^u) = \max_{c \in \mathcal{Y}_l} h_c(g(f(\mathbf{x}^u)))$, and derive soft weights:

$$w_{\text{old}}(\mathbf{x}^u) = s_{ID}(g(f(\mathbf{x}^u))), \quad w_{\text{new}}(\mathbf{x}^u) = 1 - s_{ID}(g(f(\mathbf{x}^u))), \tag{5}$$

where w_{old} indicates the likelihood of belonging to known classes and w_{new} to novel classes. These weights guide our bidirectional knowledge transfer: samples with high w_{old} participate in semantic pattern transfer from labeled data, while samples with high w_{new} contribute to relational consistency learning for novel discovery.

3.2 Relational Pattern Consistency (RPC)

Unlike prior GCD methods that treat labeled and unlabeled data as two disjoint domains, RPC explicitly couples them through bidirectional knowledge transfer, as shown in Fig. 1. Our approach

rests on two key insights: (i) labeled data contains rich semantic structure that can guide unlabeled known-class learning beyond unreliable pseudo-labels, and (ii) samples from the same category exhibit consistent behavioral signals when interacting with known-class prototypes, and critically, labeled data provides the reference framework to capture and interpret these behavioral patterns for category discovery.

3.2.1 Known-Class Preservation via Semantic Transfer. For known-class preservation, we transfer robust semantic patterns from labeled to unlabeled data through behavioral consistency under augmentation. We construct training batches by selecting ID-likely unlabeled samples based on OVA scores. For each batch of $\mathcal{B}_l$ labeled samples, we first sample μ unlabeled samples and compute $\mu_{ID} = \lfloor \mu \cdot \rho_{ID} \rfloor$, where ρ_{ID} is the estimated proportion of ID-class samples in the unlabeled set. We then select the top-μ_{ID} samples with highest w_{old} scores from these μ samples, resulting in the pattern: $[\mathbf{x}_1^l, \mathbf{x}_{1,1}^u, \ldots, \mathbf{x}_{1,\mu_{ID}}^u, \mathbf{x}_2^l, \mathbf{x}_{2,1}^u, \ldots, \mathbf{x}_{2,\mu_{ID}}^u, \ldots]$, where $\mathbf{x}_{i,j}^u$ denotes the j-th ID-class unlabeled sample paired with the i-th labeled sample, all sharing identical augmentations.

To enable semantic transfer from labeled to unlabeled samples, we apply confidence-weighted embedding [?]. Given the stacked batch embeddings $\mathbf{Z} \in \mathbb{R}^{Q \times d}$ where $Q = |\mathcal{B}_l|(1 + \mu_{ID})$, the fused embeddings $\mathbf{Z}' \in \mathbb{R}^{Q \times d}$ are obtained by:

$$\mathbf{Z}' = (\mathbf{I} + \mathbf{A})\mathbf{Z} \tag{6}$$

where $\mathbf{I} \in \mathbb{R}^{Q \times Q}$ is the identity matrix and $\mathbf{A}$ is the fusion matrix defined as follows. Following the circular shift design:

$$A_{ij} = \begin{cases} \alpha \cdot w_{\text{old}}(\mathbf{x}_i) & \text{if } j = (i-1) \bmod Q \\ -\alpha \cdot w_{\text{old}}(\mathbf{x}_i) & \text{if } i = j \\ 0 & \text{otherwise} \end{cases} \tag{7}$$

This design ensures that each unlabeled sample $\mathbf{x}_{k,m}^u$ (the m-th unlabeled sample paired with the k-th labeled sample) receives information from its preceding sample in the batch, which is either the paired labeled sample (when $m = 1$) or another unlabeled sample. The weighting by w_{old} ensures that only high-confidence ID samples are significantly affected by the fusion.

With these fused embeddings $\mathbf{Z}'$, we enforce behavioral consistency between each labeled sample and its paired unlabeled samples:

$$\mathcal{L}_{\text{align}} = \frac{1}{|\mathcal{B}_l|} \sum_{i=1}^{|\mathcal{B}_l|} \left\| \Delta(\mathbf{x}_i^l) - \frac{\sum_{j=1}^{\mu_{ID}} w_{\text{old}}(\mathbf{x}_{ij}^u) \cdot \Delta'(\mathbf{x}_{ij}^u)}{\sum_{j=1}^{\mu_{ID}} w_{\text{old}}(\mathbf{x}_{ij}^u)} \right\|_2^2, \tag{8}$$

where $\Delta(\mathbf{x}) = g(f(\mathcal{T}_w(\mathbf{x}))) - g(f(\mathcal{T}_s(\mathbf{x})))$ captures augmentation-induced behavioral changes, with $\mathcal{T}_w$ and $\mathcal{T}_s$ denoting weak and strong augmentations respectively. For labeled samples, $\Delta(\mathbf{x}^l)$ is computed on original features, while for unlabeled samples, $\Delta'(\mathbf{x}^u)$ is computed on fused features from $\mathbf{Z}'$.

3.2.2 Category Discovery via Relational Matching. For category discovery in this step, we leverage a key insight: while novel classes are semantically distinct from known classes, samples from the same novel category usually share consistent patterns of similarity and dissimilarity with the known-class reference system. Specifically, if two samples belong to the same novel class, they should exhibit similar distances to each known-class prototype—being similarly far from some known classes and similarly close to others. This shared relational pattern serves as an implicit signature of category membership.

To operationalize this, we encode each sample's relational signature as its similarity pattern with all known-class prototypes:

$$\mathbf{r}(\mathbf{x}) = \left[\frac{f(\mathbf{x}) \cdot \mathbf{p}_1}{\|f(\mathbf{x})\|_2 \|\mathbf{p}_1\|_2}, \ldots, \frac{f(\mathbf{x}) \cdot \mathbf{p}_{C_L}}{\|f(\mathbf{x})\|_2 \|\mathbf{p}_{C_L}\|_2} \right] \in \mathbb{R}^{C_L}, \tag{9}$$

where $\{\mathbf{p}_c\}_{c=1}^{C_L}$ are known-class prototypes. This transforms each sample into coordinates within the known-class reference frame. We enforce relational consistency among feature-similar samples:

$$\mathcal{L}_{\text{new}} = \sum_{i,j \in \mathcal{B}_u} w_{\text{new}}(\mathbf{x}_i^u) \cdot w_{\text{new}}(\mathbf{x}_j^u) \cdot s_{ij} \cdot \|\mathbf{r}(\mathbf{x}_i^u) - \mathbf{r}(\mathbf{x}_j^u)\|_2^2, \tag{10}$$

where $s_{ij} = \exp(\langle f(\mathbf{x}_i^u), f(\mathbf{x}_j^u) \rangle / (\|f(\mathbf{x}_i^u)\|_2 \|f(\mathbf{x}_j^u)\|_2 \cdot \tau_u))$ weights pairs by feature similarity. This transforms novel discovery from ill-posed clustering into well-defined pattern matching—samples from the same novel category share consistent relationships with all known classes.

3.3 Training Objective

The complete RPC objective integrates baseline learning with our bidirectional consistency mechanisms:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{baseline}} + \lambda_1 \mathcal{L}_{\text{align}} + \lambda_2 \mathcal{L}_{\text{new}} \tag{11}$$

where λ_1 and λ_2 balance known-class preservation and novel class relational consistency respectively.

4 Experiments

Datasets. We conduct experiments on six benchmarks covering diverse scenarios. Three fine-grained datasets from the Semantic Shift Benchmark (SSB)—CUB [?], Stanford Cars [?], and FGVC-Aircraft [?]—are used to evaluate performance on specialized domains. Two general benchmarks, CIFAR10/100 [?] and ImageNet-100 [?], assess broader object recognition capabilities. Consistent with the GCD setting [?], we split categories into known ($\mathcal{Y}_l$) and novel classes, where 50% of known-class images form the labeled data $\mathcal{D}_L$ and all other images constitute the unlabeled pool $\mathcal{D}_U$. See Tab. 1 for the specific known/novel split per dataset; the total category count K ranges from 10 (CIFAR10) to 200 (CUB).

Implementation details. Following SimGCD [?], we adopt a ViT-B/16 backbone pre-trained with DINO [?], consistent with the set up in [?]. Image features, $f(\cdot)$, are extracted from the 768-dimensional [CLS] token. We fine-tune only the final block with a batch size of 128 over 200 epochs, starting with a learning rate of 0.1 and applying cosine learning rate decay. Following [?], the balance factor λ is 0.35, and the temperature parameters τ_s and τ_u are set to 0.1 and 0.07, respectively. In this study, we empirically set $\lambda_1 = 0.5$, $\lambda_2 = 0.3$, and the embedding fusion weight $\alpha = 0.3$ based on hyperparameter analysis.

Compared methods. The compared methods include three competing novel category discovery (NCD) methods RS+ [?], UNO+ [?], and k-means [?] with DINO [?] features, and several state-of-the-art generalized category discovery (GCD) methods, including GCD [?], PromptCAL [?], DCCL [?], SimGCD [?], SPTNet [?], ProtoGCD [?] and DebGCD [?]. Importantly, Our method builds on

Table 1: Comparison with state-of-the-art methods on generic and fine-grained benchmarks. All values are percentages.

	CUB			Stanford Cars			FGVC-Aircraft			CIFAR10			CIFAR100			ImageNet-100		
Methods	All	Old	New	All	Old	New	All	Old	New	All	Old	New	All	Old	New	All	Old	New
k-means [?]	34.3	38.9	32.1	12.8	10.6	13.8	16.0	14.4	16.8	83.6	85.7	82.5	52.0	52.2	50.8	72.7	75.5	71.3
RS+ [?]	33.3	51.6	24.2	28.3	61.8	12.1	26.9	36.4	22.2	46.8	19.2	60.5	58.2	77.6	19.3	37.1	61.6	24.8
UNO+ [?]	35.1	49.0	28.1	35.5	70.5	18.6	40.3	56.4	32.2	68.6	98.3	53.8	69.5	80.6	47.2	70.3	95.0	57.9
GCD [?]	51.3	56.6	48.7	39.0	57.6	29.9	45.0	41.1	46.9	91.5	97.9	88.2	73.0	76.2	66.5	74.1	89.8	66.3
PromptCAL [?]	62.9	64.4	62.1	50.2	70.1	40.6	52.2	52.2	52.3	97.9	96.6	98.5	81.2	84.2	75.3	83.1	92.7	78.3
DCCL [?]	63.5	60.8	64.9	43.1	55.7	36.2	-	-	-	96.3	96.5	96.9	75.3	76.8	70.2	80.5	90.5	76.2
SimGCD [?]	60.3	65.6	57.7	53.8	71.9	45.0	54.2	59.1	51.8	97.1	95.1	98.1	80.1	81.2	77.8	83.0	93.1	77.9
SPTNet [?]	65.8	68.8	65.1	59.0	79.2	49.3	59.3	61.8	58.1	97.3	95.0	98.6	81.3	84.3	75.6	85.4	93.2	81.4
ProtoGCD [?]	63.2	68.5	60.5	53.8	73.7	44.2	56.8	62.5	53.9	97.3	95.3	98.2	81.9	82.9	80.0	84.0	92.2	79.9
DebGCD [?]	66.3	71.8	63.5	65.3	81.6	57.4	61.7	63.9	60.6	97.2	94.8	98.4	83.0	84.6	79.9	85.9	94.3	81.6
RPC (Ours)	**67.1**	**73.2**	**64.8**	**65.5**	**81.0**	**57.8**	**62.1**	**65.5**	**61.0**	**97.6**	**95.2**	**98.8**	**83.9**	**85.1**	**81.0**	**86.2**	93.8	**82.1**

SimGCD's [?] framework and DebGCD's [?] OVA decomposition, adding bidirectional knowledge transfer.

4.1 Results and Discussion

Results analysis. Tab. 1 shows RPC achieves state-of-the-art performance across most benchmarks. Three key findings emerge: (1) Old ACC improves significantly (73.2% on CUB compared to DebGCD's 71.8%), validating our semantic transfer mechanism. (2) New ACC gain 1.3% on CUB and 0.5% on ImageNet-100, showing relational consistency's effectiveness. (3) Fine-grained datasets yield larger overall improvements (0.8% on CUB, 0.4% on Aircraft) than generic ones (0.3% on ImageNet-100), indicating relational patterns better capture subtle distinctions.

Old-New trade-off. On ImageNet-100 and Stanford Cars, RPC improves New ACC while Old ACC is slightly lower than DebGCD (93.8% vs. 94.3% and 81.0% vs. 81.6%, respectively). We attribute this to the embedding fusion mechanism, which introduces mild perturbation to known-class decision boundaries in order to better accommodate novel-class representations. On the three SSB benchmarks, where relational patterns carry richer semantic cues, both Old and New ACC improve simultaneously.

Effect of class composition. The evaluation covers novel-class ratios from 20% (CIFAR100: 20 out of 100 classes) to 50% (CUB, Stanford Cars, FGVC-Aircraft, CIFAR10, and ImageNet-100). RPC shows consistent gains under both regimes: +1.1% New ACC on CIFAR100 and +1.3% on CUB over DebGCD. Following [? ?], we fix the labeled ratio at 50% of known-class images throughout.

Ablation Study Tab. 2 validates each component's contribution. Removing embedding fusion or $\mathcal{L}_{\text{align}}$ reduces known-class accuracy by 3.1% and 3.2% on CUB respectively, confirming their role in semantic transfer. Without $\mathcal{L}_{\text{new}}$, novel-class accuracy drops 3.3% on CUB, demonstrating its effectiveness for category discovery. All components prove essential for optimal performance.

Hyperparameter Analysis. Fig. 2 shows RPC's robustness on CUB. Optimal values ($\lambda_1 = 0.5$, $\lambda_2 = 0.3$, $\alpha = 0.3$) balance known-class preservation, category discovery, and feature fusion. Performance degrades gracefully beyond these points—excessive λ_1 over-constrains adaptation, high λ_2 causes unstable classification, and large α leads to over-fusion. Overall variation remains within 1%, validating our method's stability.

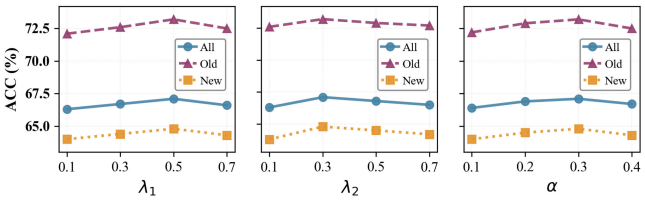

Figure 2: Impact of hyperparameters λ_1, λ_2, and α on CUB.

Table 2: Ablation study on CUB and ImageNet-100.

	CUB			ImageNet-100		
Methods	All	Old	New	All	Old	New
w/o Embedding Fusion	64.9	70.1	62.4	84.0	90.6	81.2
w/o $\mathcal{L}_{\text{align}}$	65.6	71.0	63.6	84.6	91.4	81.5
w/o $\mathcal{L}_{\text{discover}}$	65.2	73.0	61.5	84.3	92.6	79.0
RPC (Ours)	**67.1**	**73.2**	**64.8**	**86.2**	**93.8**	**82.1**

Table 3: Computational overhead comparison on CUB dataset.

Method	Params (M)	FLOPs (G)
SimGCD	92.25	17.60
RPC (Ours)	93.54 (+1.40%)	17.92 (+1.82%)

Computational Efficiency Analysis. To evaluate the practical applicability of the proposed RPC, we compare its computational overhead with the baseline SimGCD. As summarized in Table 3, RPC introduces a minimal increase in model complexity. Specifically, our method only adds 1.29M parameters (a 1.40% increase) and 0.32G FLOPs (a 1.82% increase) over the baseline.

Feature Visualization. To qualitatively evaluate the quality of the learned representations, we visualize the feature embeddings using t-SNE. As illustrated in Fig. 3, RPC produces a more structured and discriminative latent space compared to DebGCD. Specifically, within the areas highlighted by red boxes, it is observable that RPC effectively compresses the intra-class variance while expanding the inter-class margins. This superior separability suggests that our Relational Pattern Consistency (RPC) framework better preserves

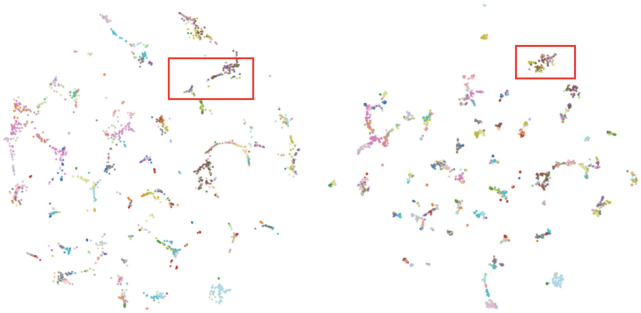

Figure 3: t-SNE visualization on CUB. (a) DebGCD and (b) RPC. Red boxes show RPC achieves tighter intra-class grouping and clearer inter-class boundaries.

the underlying semantic structure, which is crucial for identifying novel categories in a shared embedding space.

5 Conclusion

In this study, we introduced bidirectional knowledge transfer for Generalized Category Discovery (GCD) tasks. We proposed Relational Pattern Consistency (RPC), which transfers semantic patterns from labeled to unlabeled data through embedding fusion, while discovering novel categories via relational consistency with known-class prototypes. This transforms GCD from parallel learning into mutual enhancement between labeled and unlabeled data. Extensive experiments on six benchmarks demonstrate the competitiveness of the proposed RPC methods.

Following standard GCD protocols [**? ?**], RPC assumes K is known a priori. Extending RPC with automatic category number estimation [**?**] is a natural direction for future work.